01

Data engineering tasks and components

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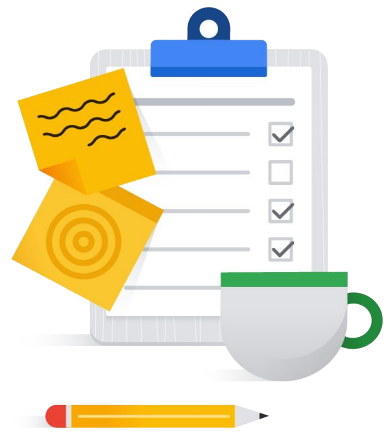
01	The role of a data engineer
02	Data sources versus data syncs
03	Data formats
04	Storage solution options on Google Cloud
05	Metadata management options on Google Cloud
06	Share datasets using Analytics Hub



In this module, first, you learn about the role of a data engineer. Second, we will cover the differences between a data source and a data sink. Then, we will review different types of data formats that a data engineer will encounter. The next topic addresses the options for storing data on Google Cloud. Then, we cover the choices available for metadata management. Finally, we will look at the features of Analytics Hub that allow you to easily share datasets both within and outside your organization.

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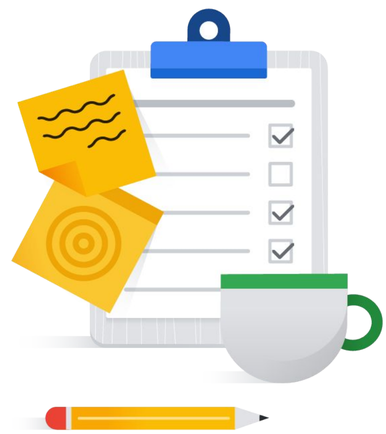
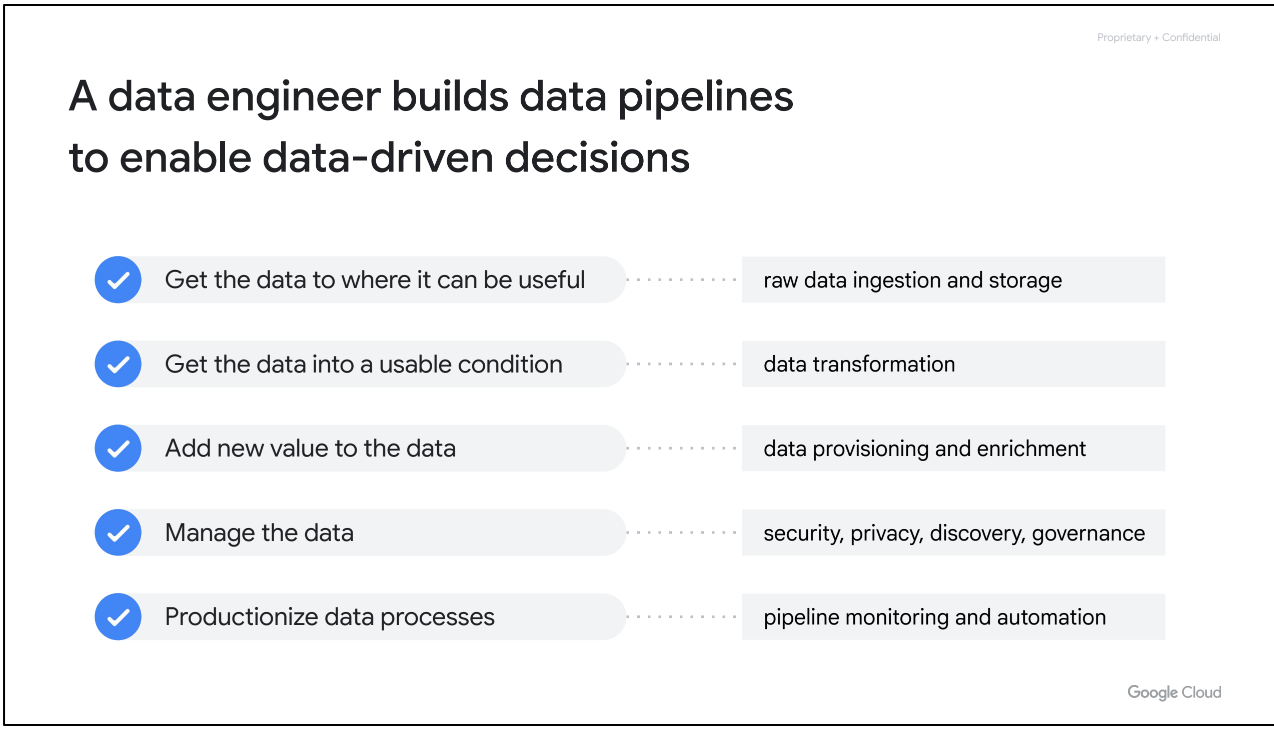


Google Cloud

Let’s start by discussing the role of a data engineer.



What does a data engineer do? At a basic level, a data engineer builds data pipelines.

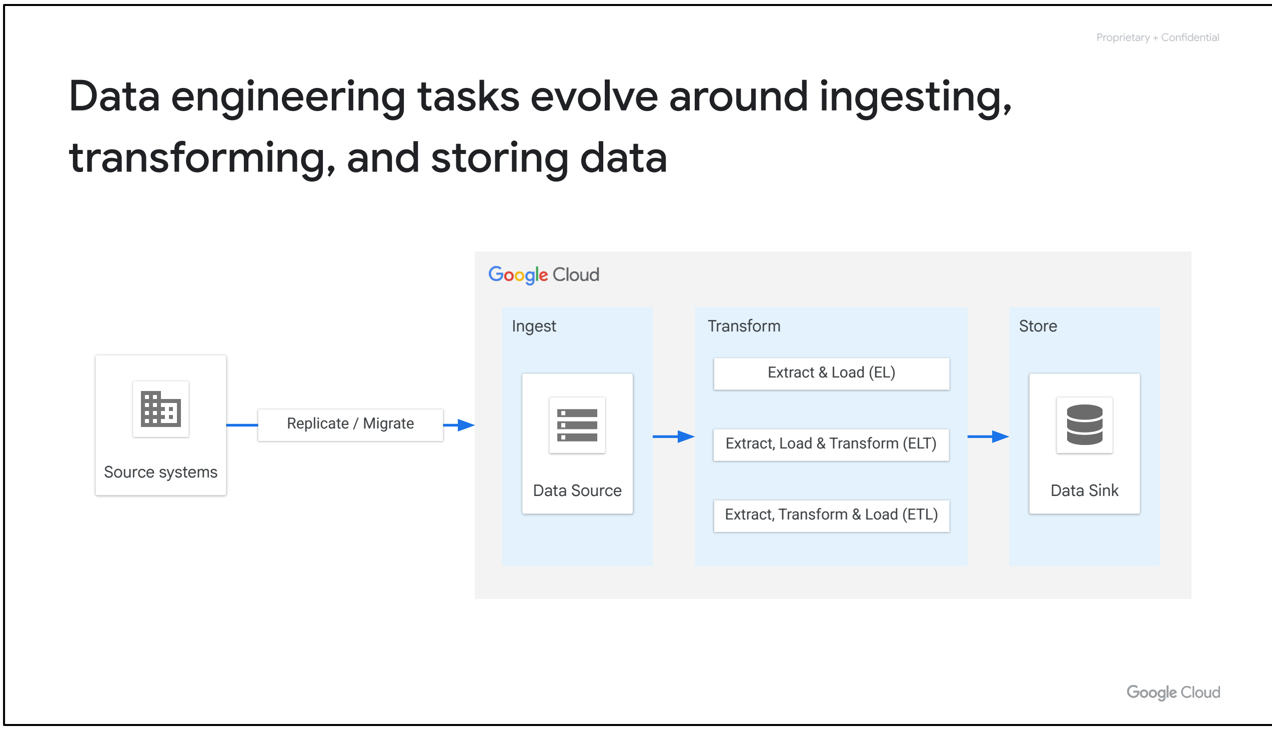
Why does the data engineer build data pipelines? Because they want to get their data into a place, such as a dashboard or report or machine learning model, from where the business can make data-driven decisions.

The data has to be in a usable condition so that someone can use this data to make decisions. Many times, the raw data is, by itself, not very useful.

Once data becomes useful, the data engineer will often apply updates or transformations to add new value to the data.

Of course, new data environments require data management practices to ensure currency and accuracy.

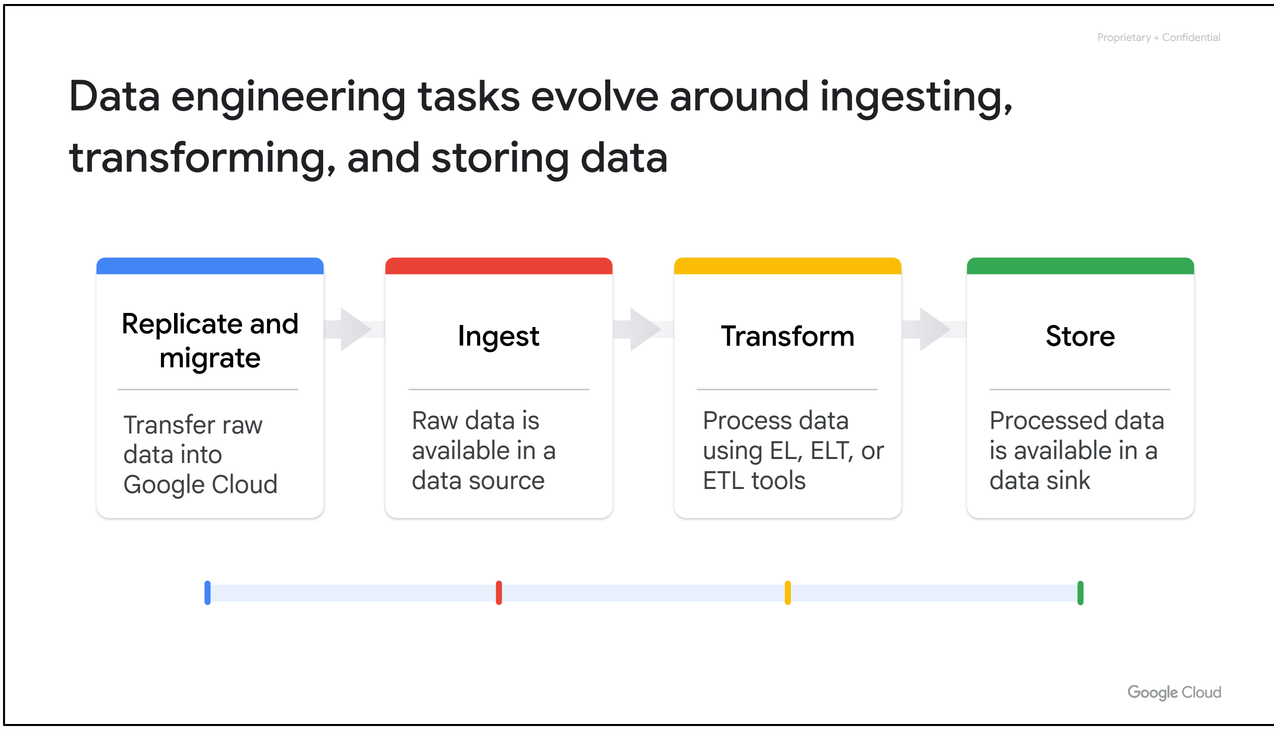
Finally, data engineers create processes and operations to move data usage into production settings.



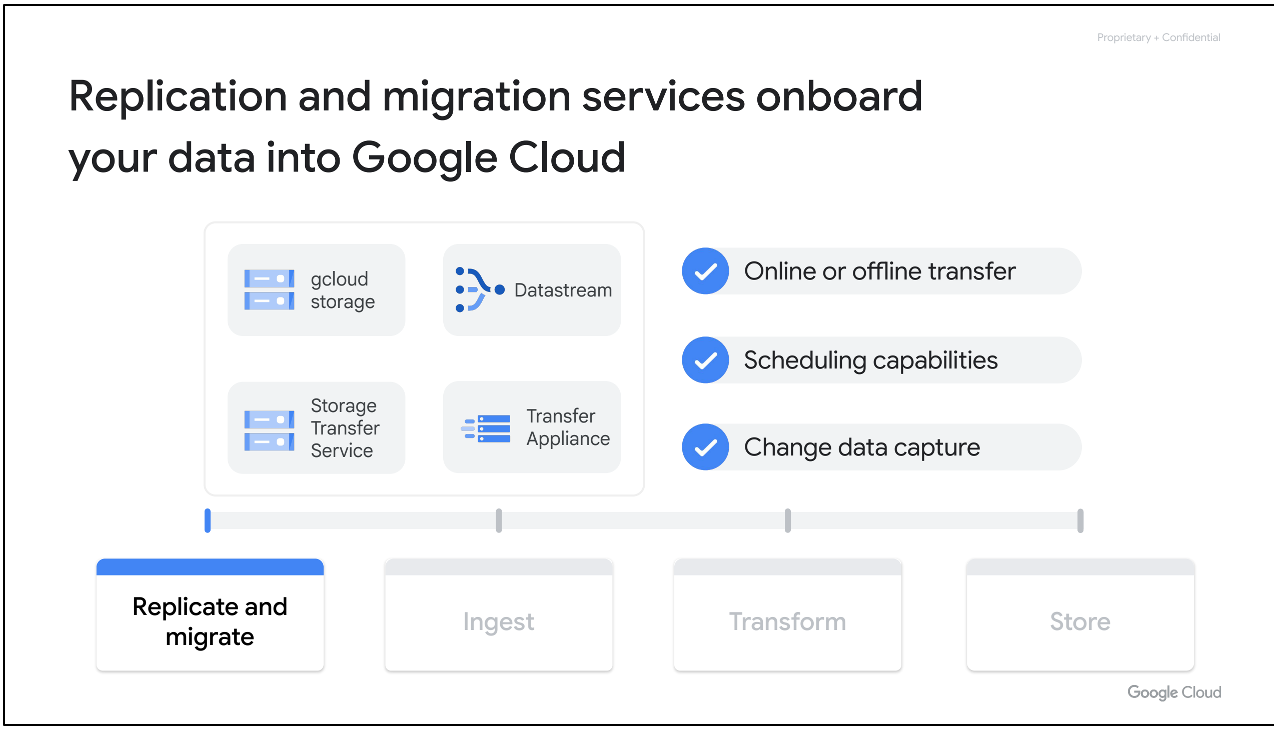
In the most basic sense, a data engineer moves data from data sources to data sinks via a transformation process.

As noted in the diagram there are generally three options or patterns to achieve those transformations.

This course focuses in detail on those three patterns.



In the most basic sense, a data engineer moves data from data sources to data sinks in four stages: replicate and migrate, ingest, transform, and store.



The replicate and migrate stage of a data pipeline focuses on the tools and options to bring data from external or internal systems into Google Cloud for further refinement.

There are a wide variety of tools and options at your disposal. They will be covered in more detail throughout this course.

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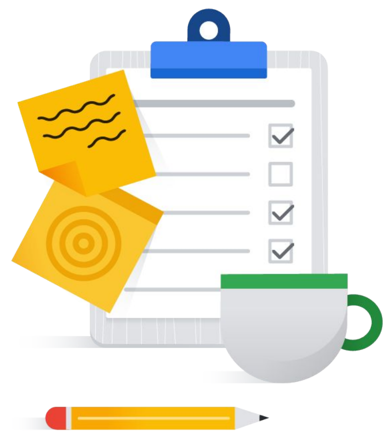
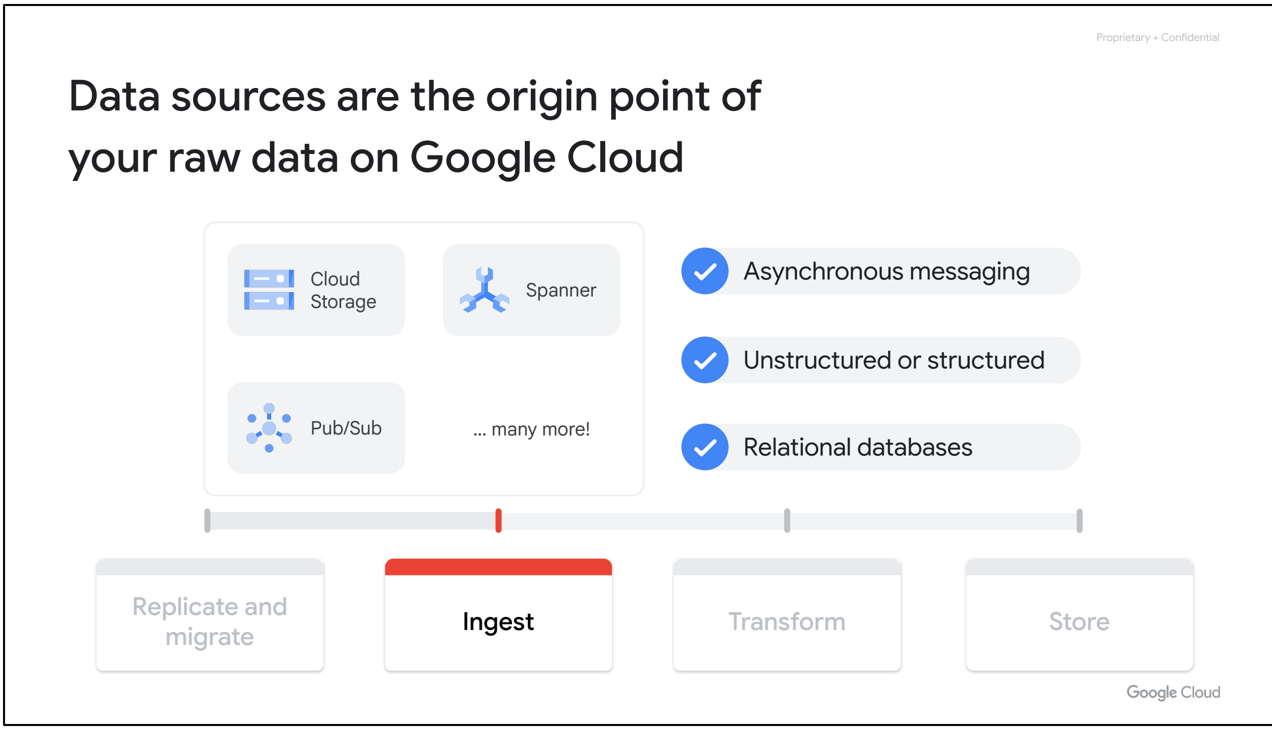
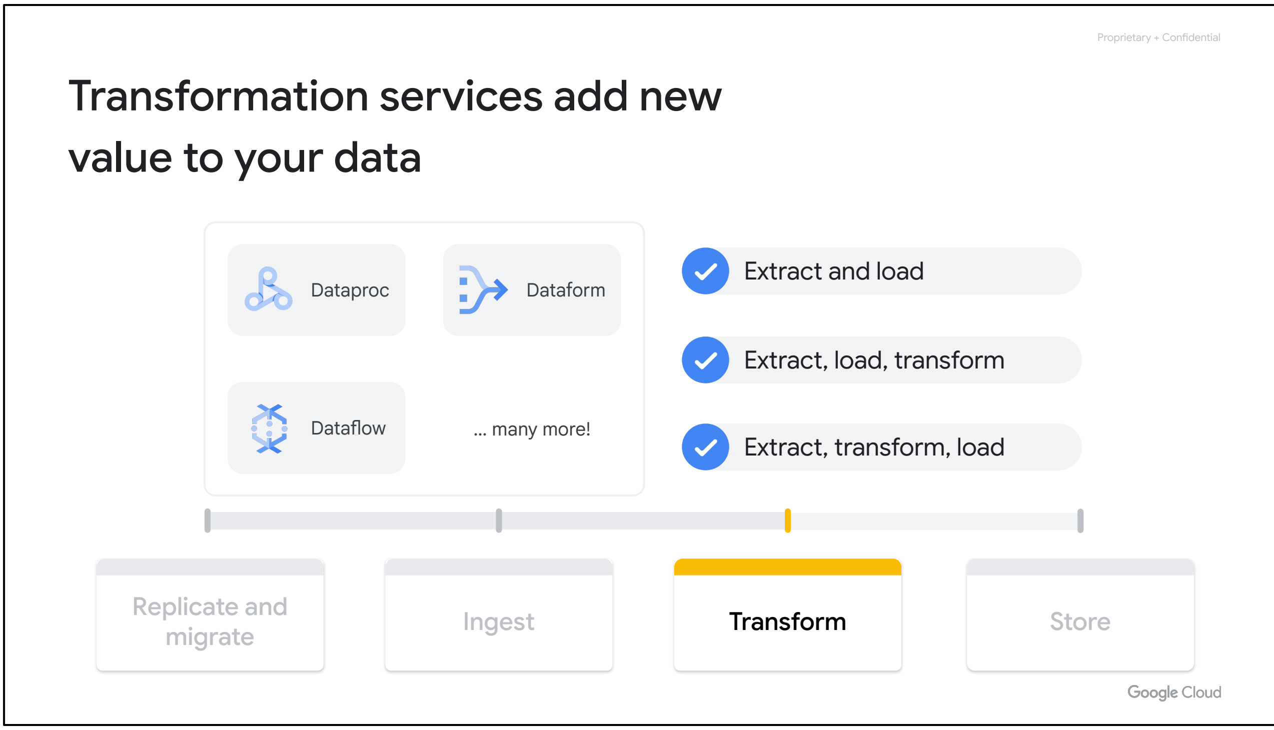




The ingest stage of a data pipeline is the point where data becomes a data source and is available for usage downstream.

Think of a data source as the starting point of your data journey. It is raw, unprocessed data waiting to be transformed into valuable insights. Any system, application, or platform that creates, stores, or shares data can be considered a data source.

Two examples of Google Cloud products used in the ingest phase are Cloud Storage, a data lake holding various types of data sources, and Pub/Sub, an asynchronous messaging system delivering data from external systems.

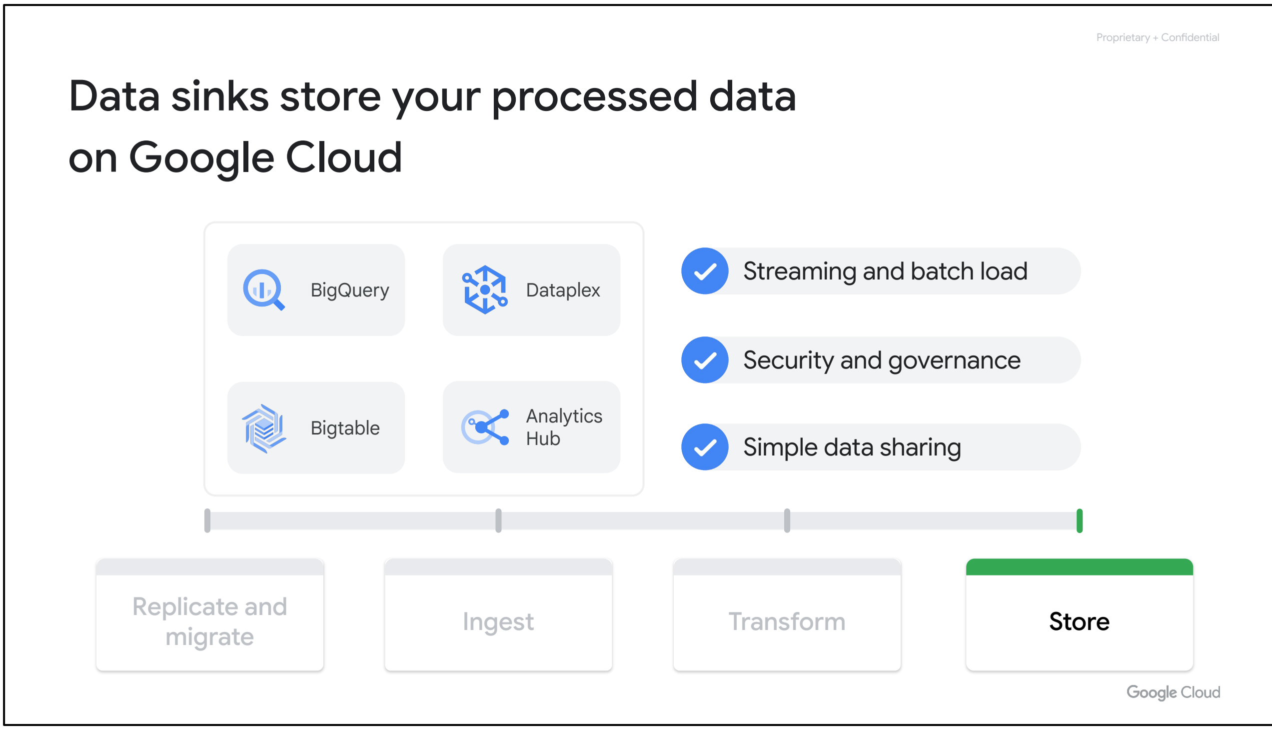


The transform stage of a data pipeline represents action taken on a data source to adjust, modify, join, or customize a data source so that it matches a specific downstream data or reporting requirement.

There are three main transformation patterns:

extract and load;
extract, load, and transform; and
extract, transform, and load.

We explore each of these patterns in their own modules later in the course.



The store stage of a data pipeline represents the last step, when we deposit data in its final form.

A data sink is the final stop in the data journey. It's where processed and transformed data is stored for future use, analysis, and decision-making. Think of it as the reservoir at the end of the river, where valuable information is collected and readily available.

Two examples of Google Cloud products used in the store phase are BigQuery, a serverless data warehouse, and Bigtable, a highly scalable NoSQL database.

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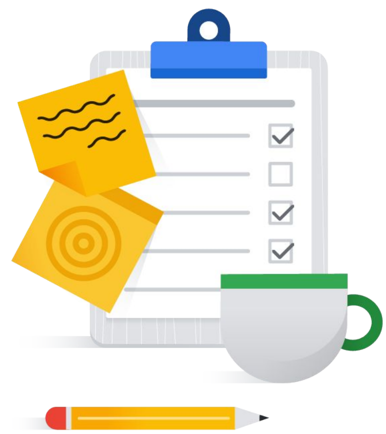

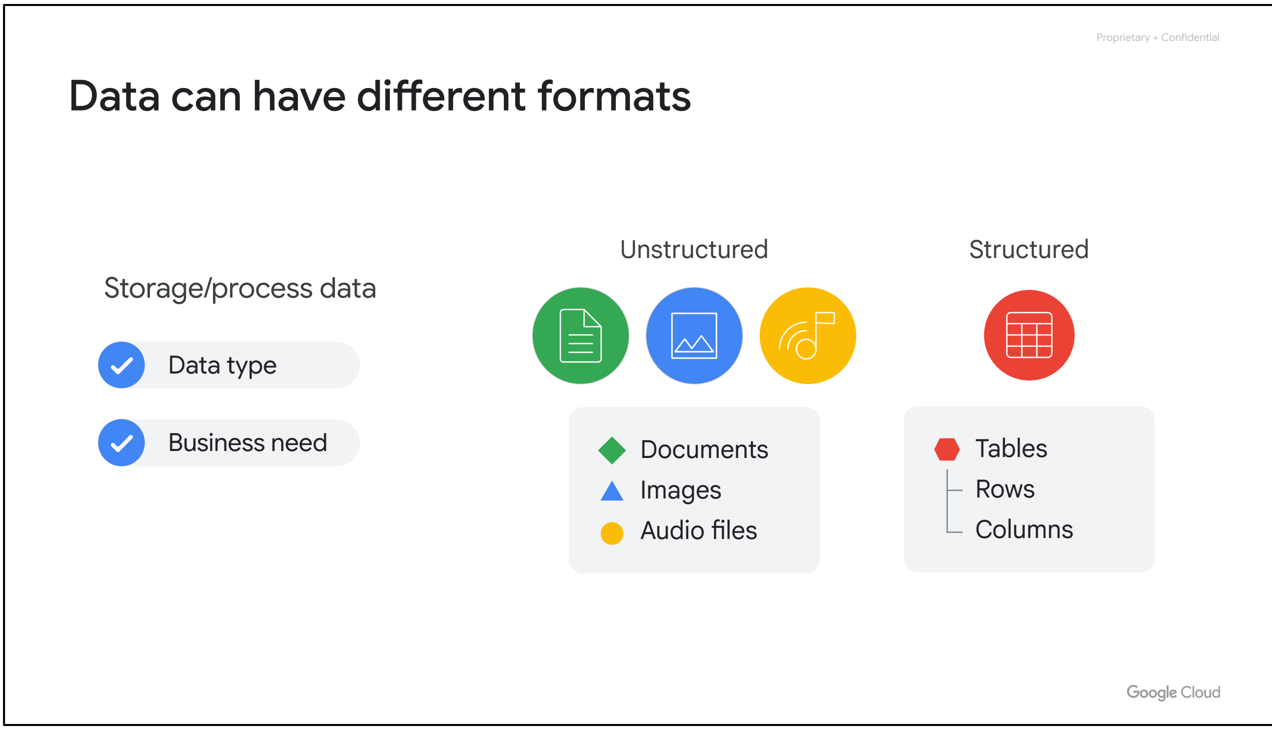

Data exists in two primary formats, unstructured and structured.

Unstructured data is information stored in a non-tabular form such as documents, images, and audio files. Unstructured data is usually suited for Cloud Storage, but BigQuery also offers the capability to store unstructured data via object tables.

There is also structured data, which represents information stored in tables, rows, and columns.

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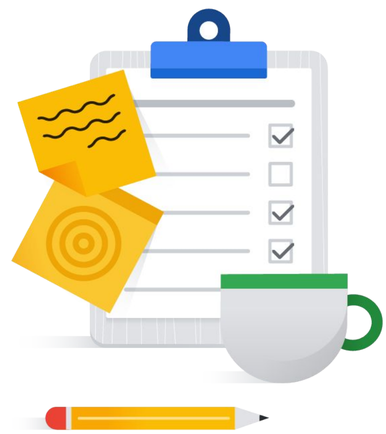
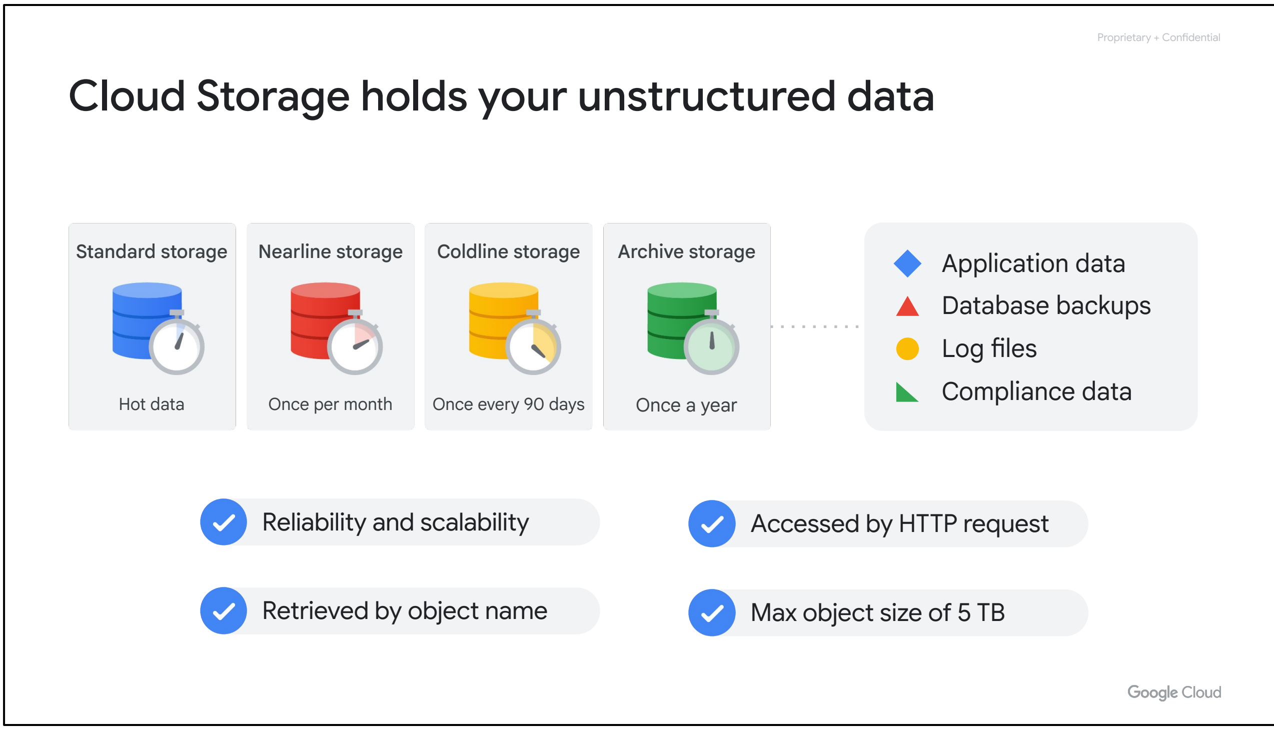


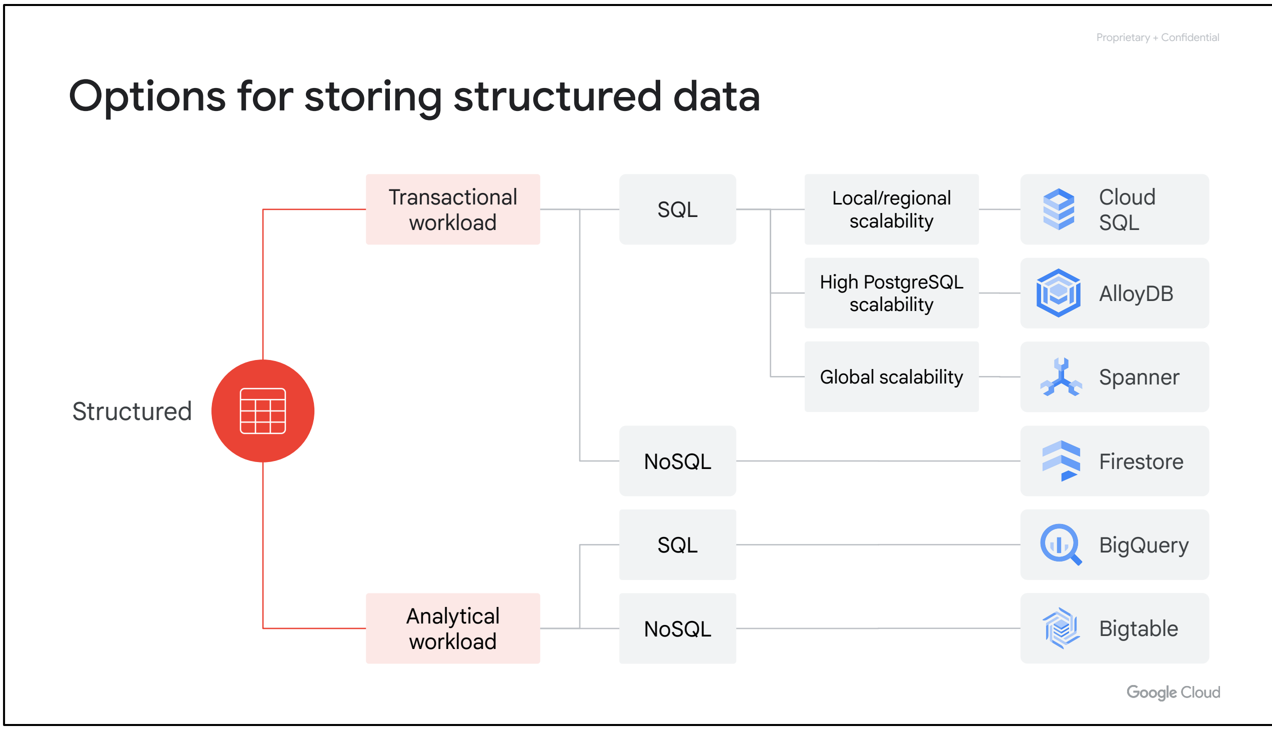


There are several key products on Google Cloud that are used by data engineers. One main product is Cloud Storage. Unstructured data is usually well-suited to be stored in Cloud Storage.

Within Cloud Storage, objects are accessed by using HTTP requests, including ranged GETs to retrieve portions of the data. The only key is the object name. There is object metadata but the object itself is treated as unstructured bytes. The scale of the system allows for serving large static content and accepting user-uploaded content including videos, photos, and files. Objects can be up to 5 Terabytes each.

Cloud Storage is built for availability, durability, scalability, and consistency. It's an ideal solution for hosting static websites and storing images, videos, objects and blobs, and any unstructured data.

Cloud Storage has four primary storage classes: standard storage, nearline storage, coldline storage, and archive storage. The classes are differentiated by the expected period of object access.



You have a full range of cost-effective storage services for structured data to choose from when developing with Google Cloud. No one size fits all, and your choice of storage and database solutions will depend on your application and workload.

Cloud SQL is Google Cloud's managed relational database service.

AlloyDB is a fully managed, high-performance PostgreSQL database service from Google Cloud.

Spanner is Google Cloud's fully managed relational database service that offers both strong consistency and horizontal scalability.

Firestore is a fast, fully managed, serverless, NoSQL document database built for automatic scaling, high performance, and ease of application development.

BigQuery is a fully managed, serverless enterprise data warehouse for analytics.

Bigtable is a high-performance NoSQL database service. Bigtable is built for fast key-value lookup and supports consistent sub 10 millisecond latency.

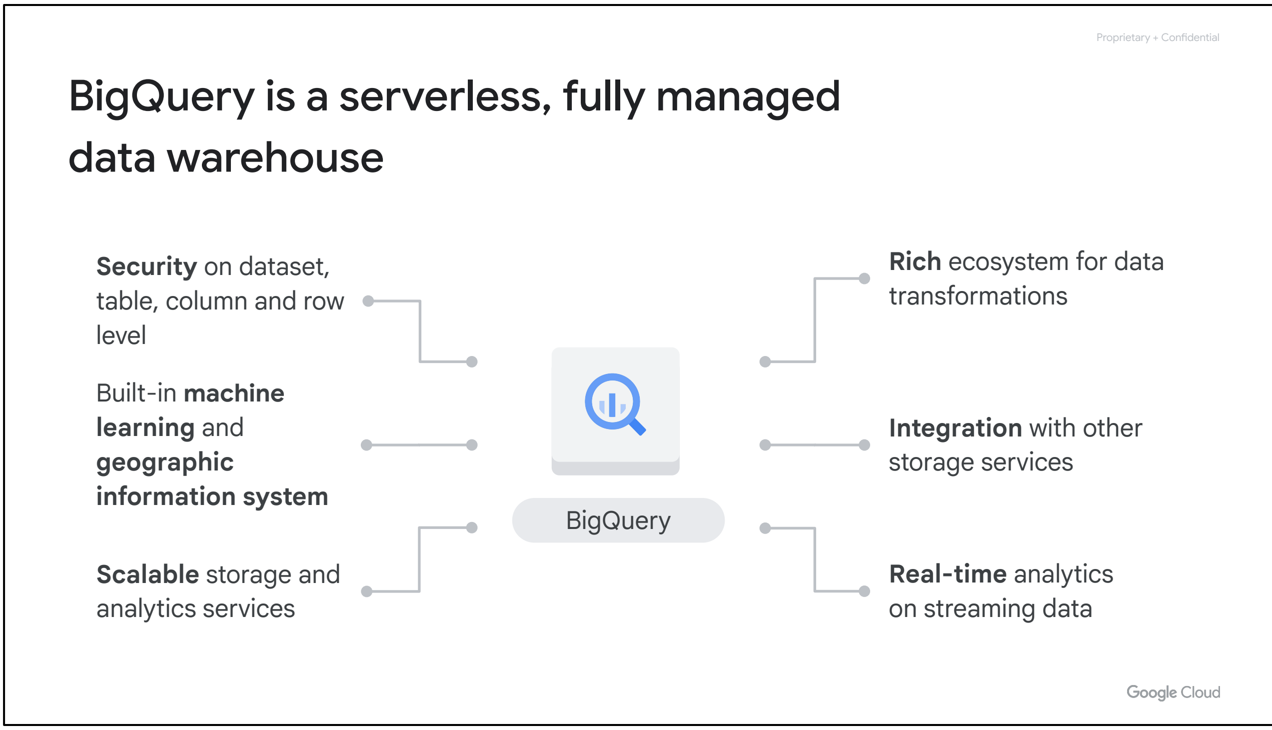
Data lake versus data warehouse

	Data lake	Data warehouse
Data format	Native (unstructured, semi-structured, structured)	Schema (structured or semi-structured)
Data type	Raw	Pre-processed and aggregated from multiple data sources
Purpose	Data science, applications, business decisions	Long-term business analysis
Dependencies	Tools/processes to enable data discovery, governance, security, metadata management	Standalone
Service	Cloud Storage	BigQuery

The two key concepts in data engineering are that of the data lake and the data warehouse.

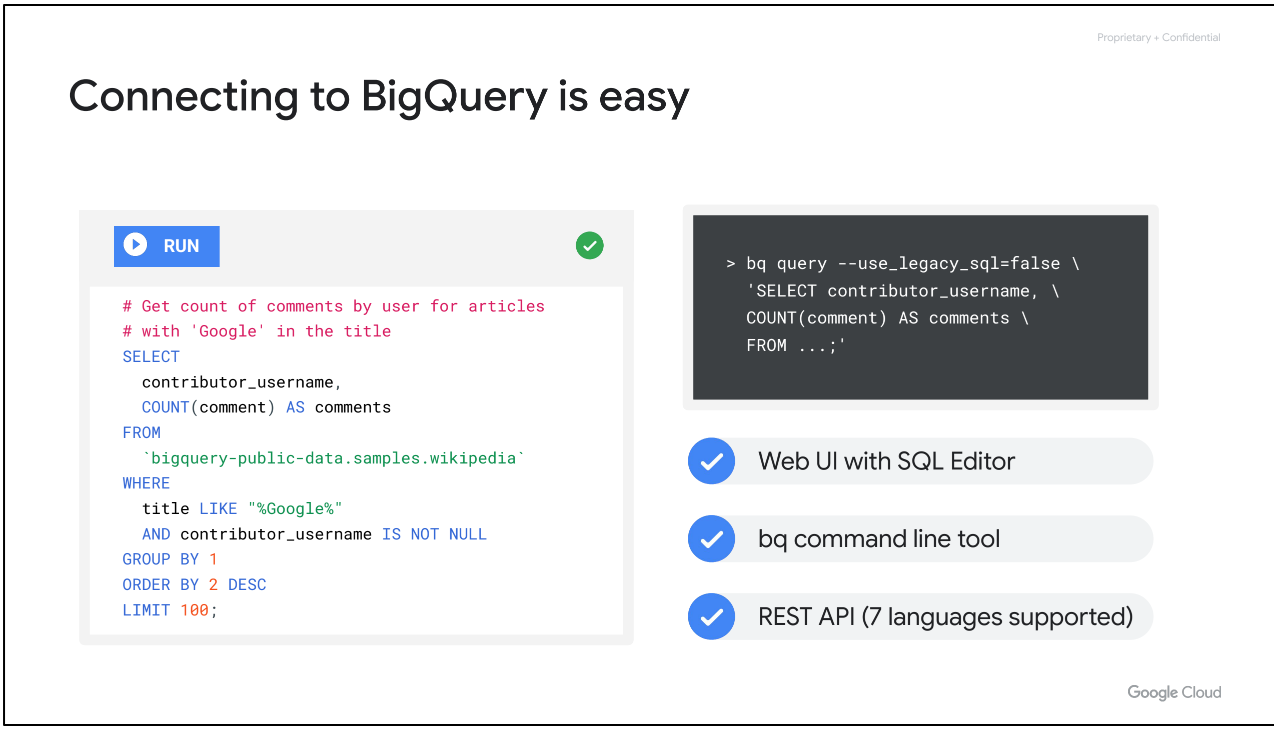
A data lake is a vast repository for storing raw, unprocessed data in various formats, including unstructured, semi-structured, and structured. It serves as a centralized storage solution for diverse data types, enabling flexible use cases like data science, applications, and business decision-making.

A data warehouse is a structured repository designed for storing pre-processed and aggregated data from multiple sources. Primarily used for long-term business analysis, it enables efficient querying and reporting for informed decision-making. Data warehouses often operate as standalone systems, independent of other data storage solutions.



BigQuery is a fully managed, serverless enterprise data warehouse for analytics. BigQuery has built-in features like machine learning, geospatial analysis, and business intelligence. BigQuery can scan terabytes in seconds, and petabytes in minutes.

BigQuery is a great solution for online analytical processing, or OLAP, workloads for big data exploration and processing. BigQuery is also well-suited for reporting with business intelligence tools.

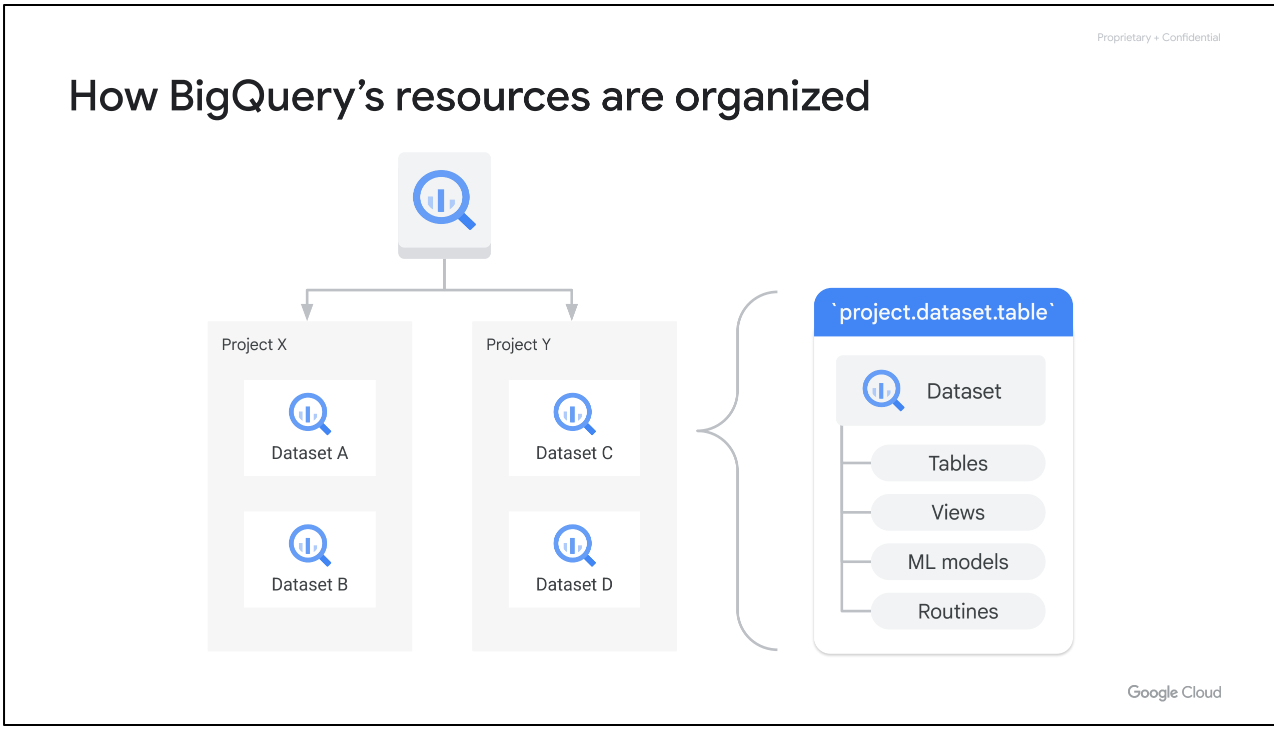


BigQuery has several easy-to-use options for accessing data.

The first is via the Google Cloud console SQL editor.

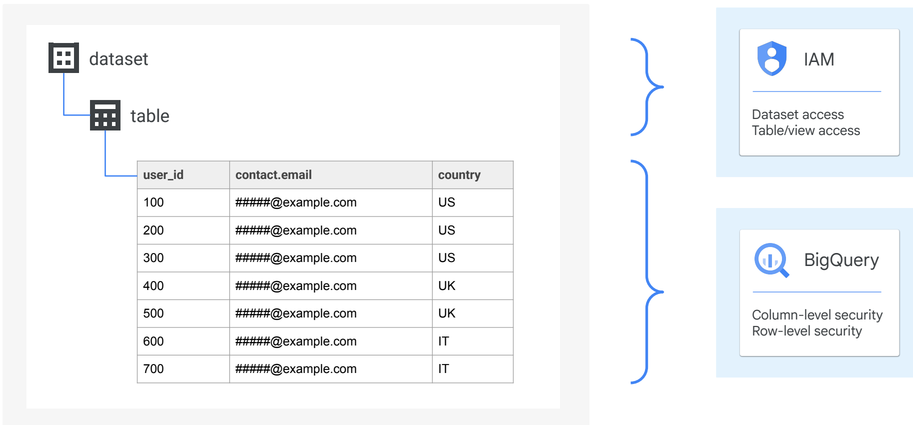
The second is via the bq command line tool which is part of the Google Cloud SDK.

The last is via a robust REST API which supports calls in seven programming languages.



BigQuery organizes data tables into units called datasets. These datasets are scoped to your Google Cloud project. When you reference a table from the command line in SQL queries or in code, you refer to it by using the construct project.dataset.table.

You can secure your BigQuery resources on multiple levels



user_id	contact.email	country
100	#####@example.com	US
200	#####@example.com	US
300	#####@example.com	US
400	#####@example.com	UK
500	#####@example.com	UK
600	#####@example.com	IT
700	#####@example.com	IT

Access control is through IAM and is at the dataset, table or view, or column level. In order to query data in a table or view, you need at least read permissions on the table or view.

Demo

Query terabytes of data in seconds

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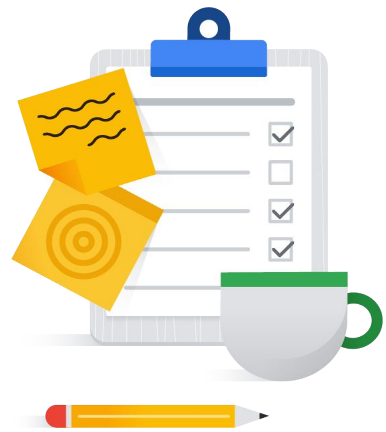
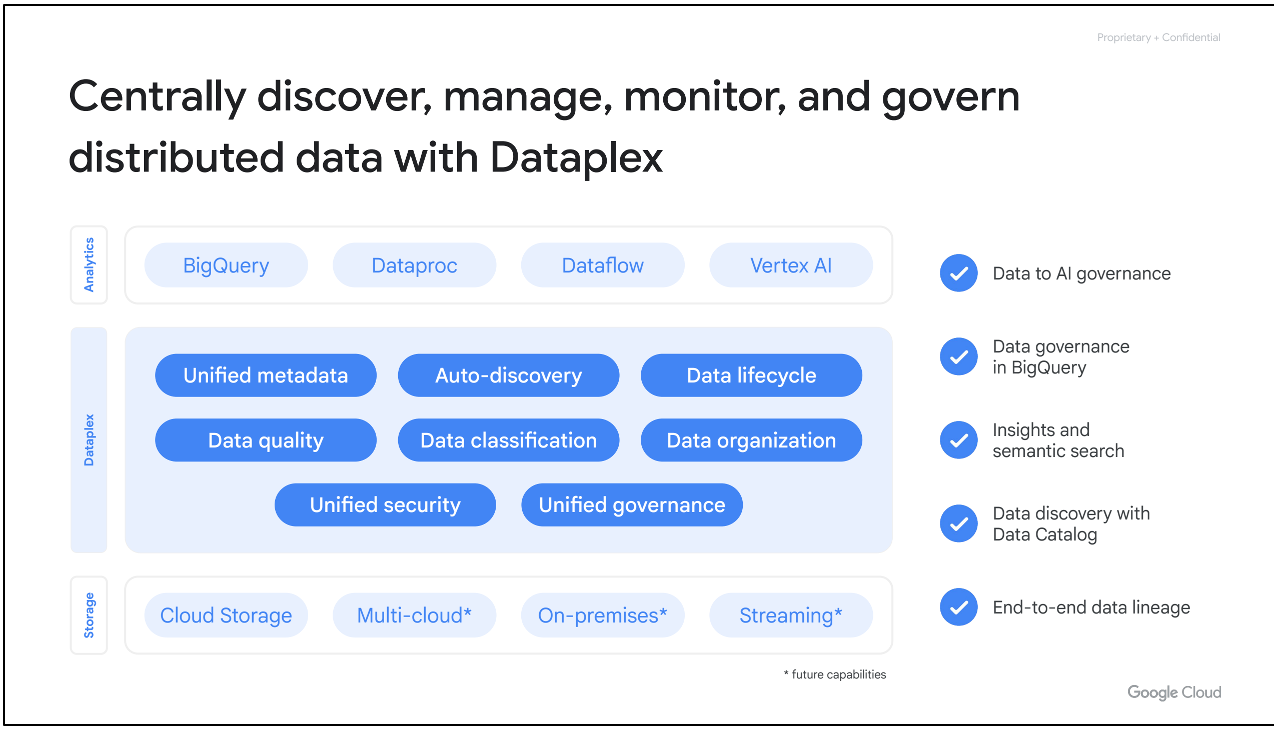




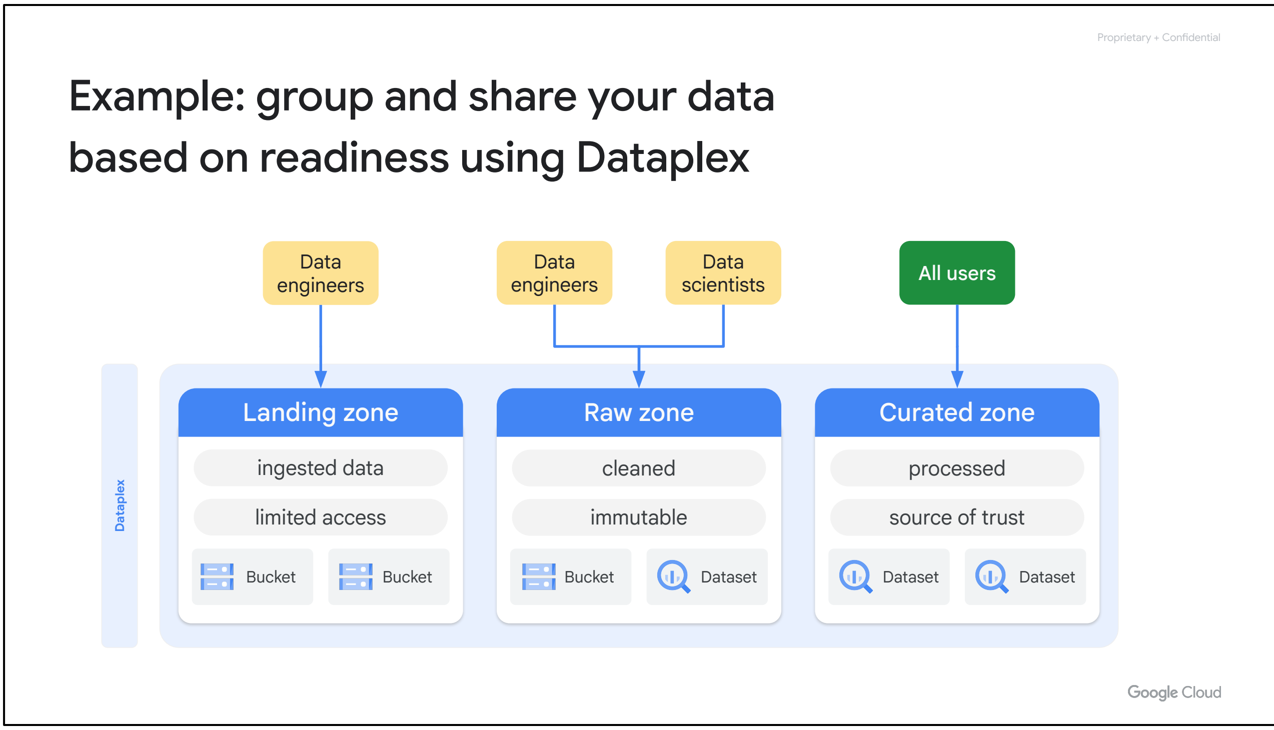
Metadata is a key element to making data more manageable and useful across an organization. Dataplex is a comprehensive data management solution that allows you to centrally discover, manage, monitor, and govern distributed data across your organization.

With Dataplex, you can break down data silos, centralize security and governance, while enabling distributed ownership, and easily search and discover data based on business context.

Dataplex also offers built-in data intelligence, support for open-source tools, and a robust partner ecosystem, helping you to trust your data and accelerate time to insights.

References:

https://cloud.google.com/dataplex
https://cloud.google.com/dataplex/docs/introduction



Dataplex lets you standardize and unify metadata, security policies, governance, classification, and data lifecycle management across this distributed data.

Another common use case is when your data is accessible only to data engineers, and is later refined and made available to data scientists and analysts. In this case, you can set up a lake to have the following:

A raw zone for the data which is accessed by data engineers and data scientists.

A curated zone for the data which is accessed by all users.

References:

https://cloud.google.com/dataplex/docs/introduction#data_tiering_based_on_readiness

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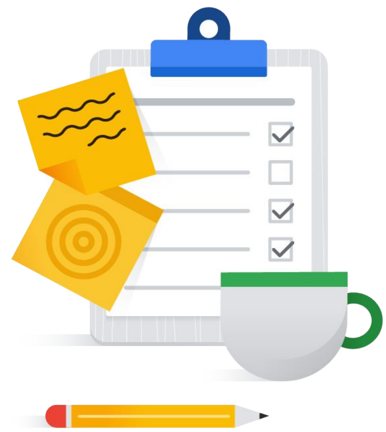
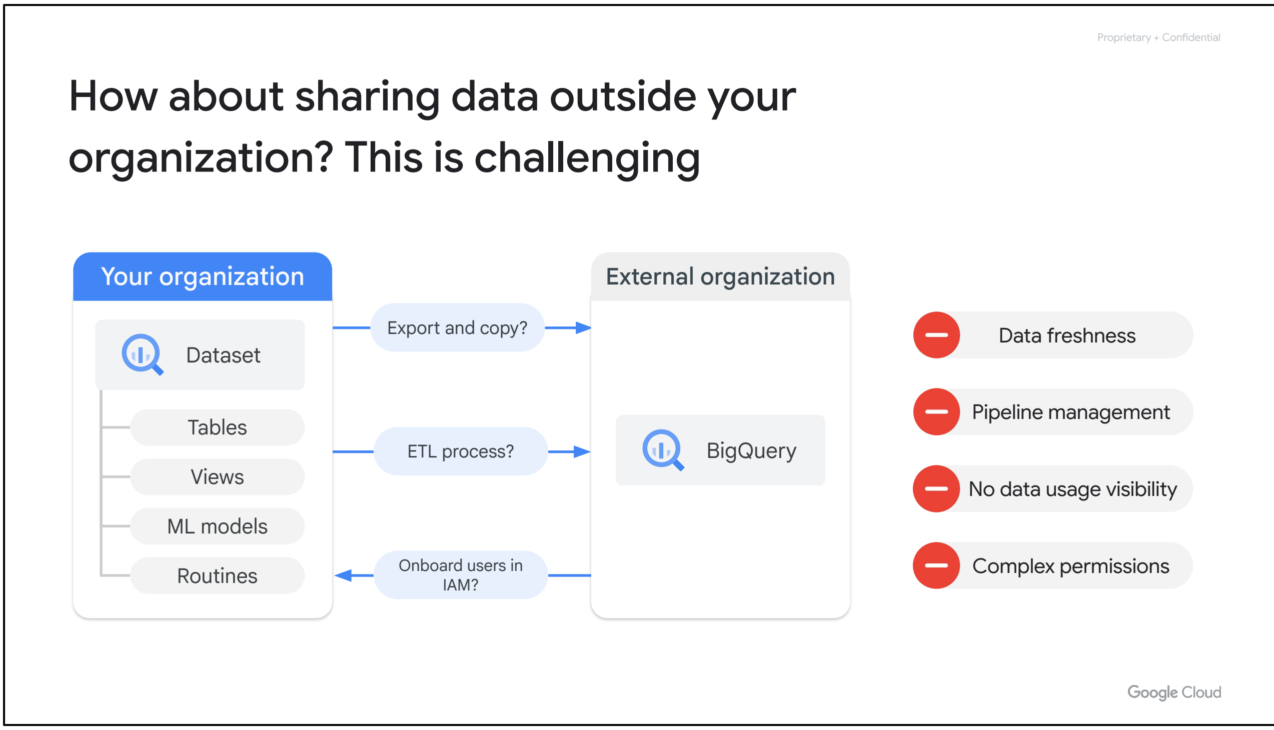
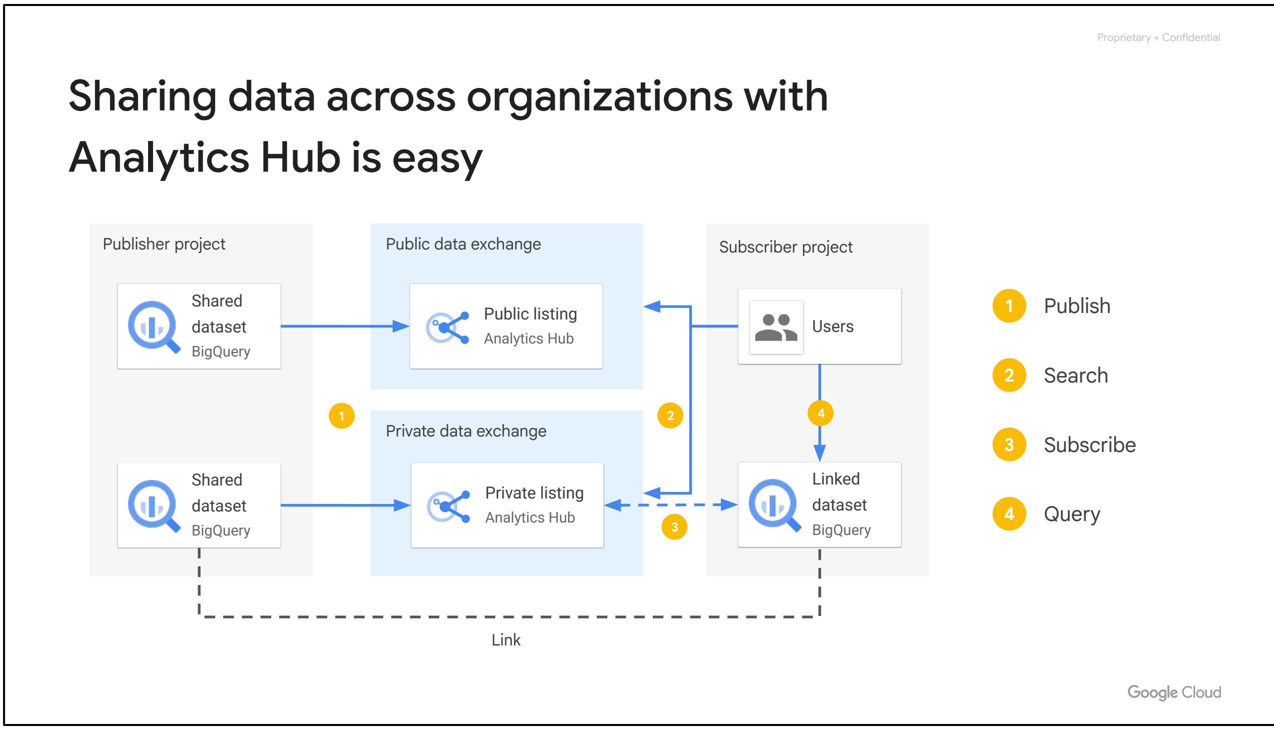




Sharing data is challenging especially outside of your organization.

You need to consider security and permissions, destination options for data pipelines, data freshness and accuracy, and finally, usage monitoring.

Analytics Hub was created to meet these data sharing challenges.



Analytics Hub helps organizations unlock the value of data sharing, leading to new insights and business value.

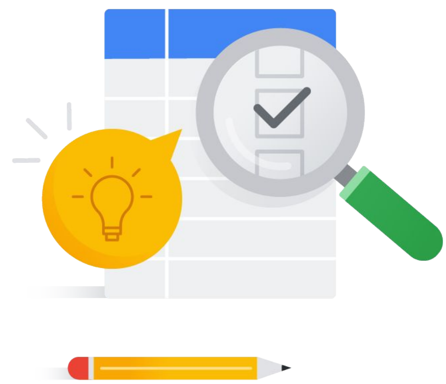
With Analytics Hub, you create a rich data ecosystem by publishing and subscribing to analytics-ready datasets. Because data is shared in place, data providers are able to control and monitor how their data is being used.

Analytics Hub provides a self-service way to access valuable and trusted data assets, including data provided by Google.

Finally, Analytics Hub provides an opportunity to monetize data assets. Analytics Hub removes the tasks of building the infrastructure required for monetization.

Lab: Loading Data into BigQuery

30 min



Learning objectives

- Load data into BigQuery from various sources.
- Load data into BigQuery using the CLI and the Google Cloud console.
- Use DDL to create tables.

In this lab, you practice loading data into BigQuery. The primary objective of this lab is to load data into BigQuery using both the command-line interface and the Google Cloud console. You also experience loading several datasets into BigQuery and using the Data Description Language or DDL.

Lab URL: https://www.cloudskillsboost.google/catalog_lab/2119

Google Cloud